

WASSIM CHOUAYAKH

+216 92 901 311 | wassimchouayakh73@gmail.com | [LinkedIn](#) | [Portfolio](#) | Tunis, Tunisia

EDUCATION

National Institute of Applied Science and Technology (INSAT) <i>Engineering Degree – Industrial IT and Automation</i> (Expected 2028)	Tunis, Tunisia Sep 2025 – Present
National Institute of Applied Science and Technology (INSAT) <i>Preparatory Cycle – Mathematics, Physics and Industrial IT</i>	Tunis, Tunisia Sep 2023 – Jun 2025

TECHNICAL SKILLS

Languages: C/C++ (bare-metal & HAL), Python, Assembly (ARM Cortex-M)
Microcontrollers: STM32 F1/F4 – HAL and register-level development, interrupts, DMA, real-time
Protocols: UART, I2C, SPI, CAN Bus – register-level driver implementation
Peripherals: GPIO, Timers, PWM (generation & input capture), ADC, EXTI
RTOS & OS: FreeRTOS (tasks, queues, semaphores), Linux/Unix, POSIX sockets, pthreads
Embedded Tools: STM32CubeIDE, STM32CubeMX, Keil uVision, ST-Link, Logic Analyzer, GDB
DevOps: Git, Docker, CMake, Make

PROJECTS

- Virtual ECU Diagnostic Stack on Linux vCAN** | *C, SocketCAN, UDS, ISO 15765-2* 2026
– Built a single-ECU diagnostic prototype supporting 5+ UDS services over Linux vCAN via SocketCAN and a C-based ISO-TP layer with under 10ms end-to-end response time
– Integrated Tester Present (0x3E) handling and negative response generation covering 10+ unsupported UDS service codes (NRC)
– Authored 12+ smoke tests covering the full ISO-TP request/response path and achieving 90%+ critical-path coverage
- Real-Time Temperature Monitoring System via CAN Bus** | *STM32 F4, C, CAN, I2C, FreeRTOS* 2025
– Architected and deployed a multi-node STM32 system with I2C sensor acquisition and bare-metal CAN Bus transmission to a central display node
– Orchestrated 4 concurrent FreeRTOS tasks (producer/consumer) for CAN Tx/Rx and ADC sampling, maintaining deterministic cycle times under 5ms
– Validated bus arbitration and timing with a Logic Analyzer; cut per-frame latency to under 1ms at 500kbps
- STM32 Peripheral & Protocol Driver Library** | *STM32 F4, C, Bare-Metal, UART, I2C, PWM, DMA* 2024
– Constructed 16+ bare-metal drivers (GPIO, Timers, PWM, ADC, UART, I2C) via direct register manipulation, bypassing any hardware abstraction layer
– Implemented PWM input capture and EXTI interrupt routines achieving sub-500ns latency across input frequencies up to 1MHz verified on a Logic Analyzer
– Wrote a full-duplex bare-metal UART driver enabling real-time bidirectional PC-to-STM32 serial communication and a register-level I2C driver for configuring and reading multiple external sensors

WORK EXPERIENCE

- Information Systems Intern – PAMS** Sep 2026 – Present
UGFS North Africa – Tunis, Tunisia
– Contributing to the information system transformation program (PAMS), helping modernize core business processes
– Collaborating with business and IT teams on the deployment and adoption of the new system
- AI Agent Developer – IaC Terraform Generator** Jul – Aug 2026
LOOYAS – Tunis, Tunisia
– LangGraph multi-agent pipeline (planner, contractor, coder, validator, security scanner, packager) generating Terraform for AWS, Azure and GCP from plain-text prompts
– Automated quality via terraform fmt/init/validate, TFLint, Checkov; deterministic HCL auto-fix + LLM refinement; stack: FastAPI, Celery, Redis, React, Docker Compose
- Software & Validation Engineering Intern** Jul – Aug 2025
KPIT Technologies (Automotive Embedded) – Sfax, Tunisia
– Streamlined the AUTOSAR test-case compliance workflow by authoring a Python automation script, reducing manual review time by ~40% and cutting report processing from hours to minutes
– Delivered a cross-platform desktop tool (Flutter + Python backend) for internal validation workflows, adopted by 10+ embedded software developers
- AI Engineering Intern** Jun – Jul 2025
Intellitech – Sfax, Tunisia
– Deployed a RAG-based chatbot (LLM + retrieval over 100+ files); ~30% better relevance, ~4h/week saved per analyst
– Automated ingestion (PDF, DOCX, TXT) via async FastAPI, 100+ documents per batch, zero manual intervention

LANGUAGES

French: Native **Arabic:** Native **English:** Fluent (professional) **Spanish:** Beginner